COURSE SLIDES · NON-WSQ

Tableau Data Prep Masterclass

Course Code: C181

Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v1 · 4 October 2026



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COURSE ADMINISTRATION

Welcome & Housekeeping

Two-day training · concept-led slides · detailed procedures in the Learner Guide

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About the Trainer



Your Trainer

Complete before class

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

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NAMED TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE
Principal trainer and curriculum developer

EXPERTISE
Data analytics, visualisation, AI and business technology

DELIVERY
Certification and applied workplace training

APPROACH
Concept clarity, realistic data and evidence-based practice

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Let's Know Each Other

1. Name, organisation and role.
2. How you currently use spreadsheets, BI or Tableau.
3. One recurring decision your data should make easier.
4. Your target outcome for the Tableau Desktop Specialist credential.

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HOUSEKEEPING

Ground Rules

1. **Participate**
   Ask, test and explain — active practice builds recall.

2. **Respect**
   One conversation at a time; critique ideas, not people.

3. **Protect data**
   Use supplied mock data; do not expose workplace records.

4. **Stay punctual**
   Return from breaks on time.

5. **Save evidence**
   Keep Tableau workbooks, screenshots and checklists.

6. **Procedures live in LG**
   Slides teach concepts; the Learner Guide gives clicks and steps.

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SCHEDULE

Lesson Plan — Day 1

Morning

- 9:30 Welcome and course orientation
- 9:45 Topic 1: connect, prepare and cleanse
- 10:50 Topic 1 labs 1–3
- 12:30 Lunch

Afternoon

- 1:00 Lab 11: Tableau Prep cleansing
- 2:10 Topic 2: charts and organisation
- 3:10 Topic 2 labs 4–7
- 5:15 Recap and Q&A
- 5:30 End

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SCHEDULE

Lesson Plan — Day 2

Morning

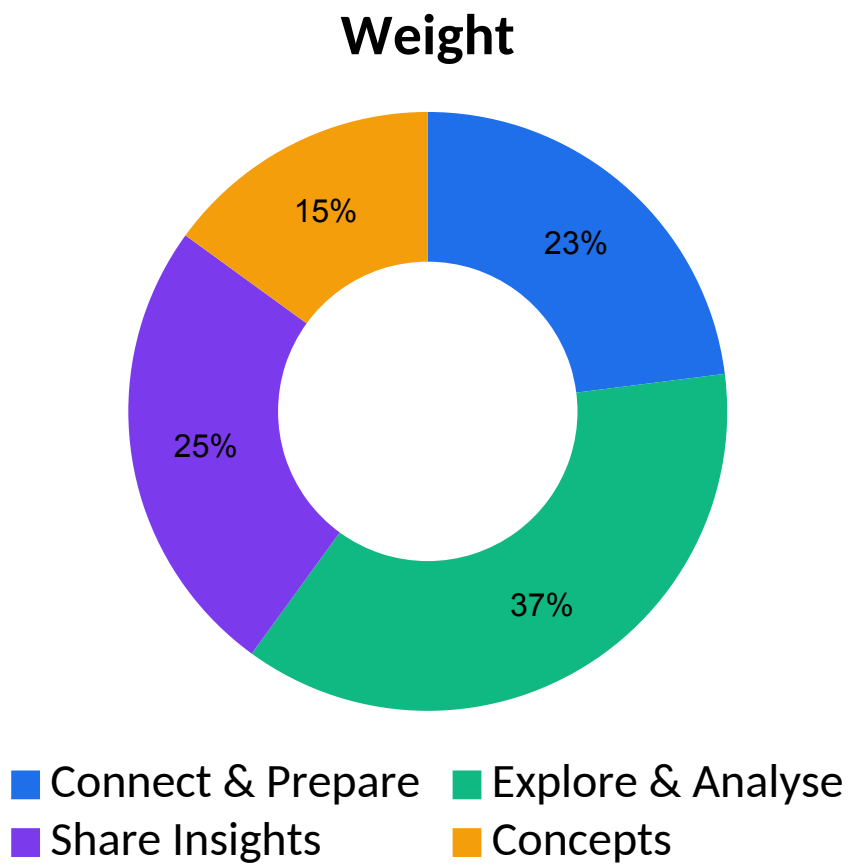
- 9:30 Recap of Day 1
- 9:50 Topic 3: format and dashboards
- 11:40 Labs 8–9
- 12:30 Lunch

Afternoon

- 1:00 Labs 8–9 continued
- 2:30 Topic 4 concepts
- 3:10 Capstone lab 10
- 5:15 Course wrap-up and feedback
- 5:30 End

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Tableau Desktop Specialist — Exam Weighting



37%
Explore and analyse receives the most teaching and practice time.

23% + 25%
Data preparation and sharing each remain substantial.

15%
Concepts are compact but foundational to every scenario.

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EXAM-ALIGNED JOURNEY

Course Outline by Official Domain

Domain	Weight	Labs
1 · Connect and Prepare	23%	Labs 1–3
2 · Explore and Analyse	37%	Labs 4–7
3 · Share Insights	25%	Labs 8–9
4 · Concepts and Preparation	15%	Lab 10

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Access the Hands-On Labs

Clone repository

- git clone the course repository
- Keeps folder structure intact
- Pull future corrections

Download ZIP

- GitHub Code → Download ZIP
- Extract before opening Excel
- Use when Git is unavailable

Every lab folder

- README with procedure
- Mock Excel workbook
- starter.twb + checklist

https://github.com/tertiarycourses/C181-Tableau-Data-Prep-Masterclass

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01

TOPIC 01

Connecting to and Preparing Data in Tableau Desktop

Official exam weighting 23% · Connections · extracts · metadata · relationships · joins · unions · field properties · Tableau Prep cleansing

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EXAM WEIGHTING 23%

Topic 01 — Core Ideas

1

Connection strategy

Live prioritises source freshness; extracts prioritise speed, portability and predictable performance.

2

Model at the right layer

Relationships preserve table grain; joins create one physical row set; unions append rows.

3

Metadata is quality

Names, aliases, types, roles, formats and defaults determine how fields behave.

4

Clean before you model

Tableau Prep Builder repairs types, splits, nulls, duplicates and inconsistent members before analysis.

5

Validate first

Check row counts, null keys, duplicated grain and unmatched records before trusting a view.

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OFFICIAL SCOPE

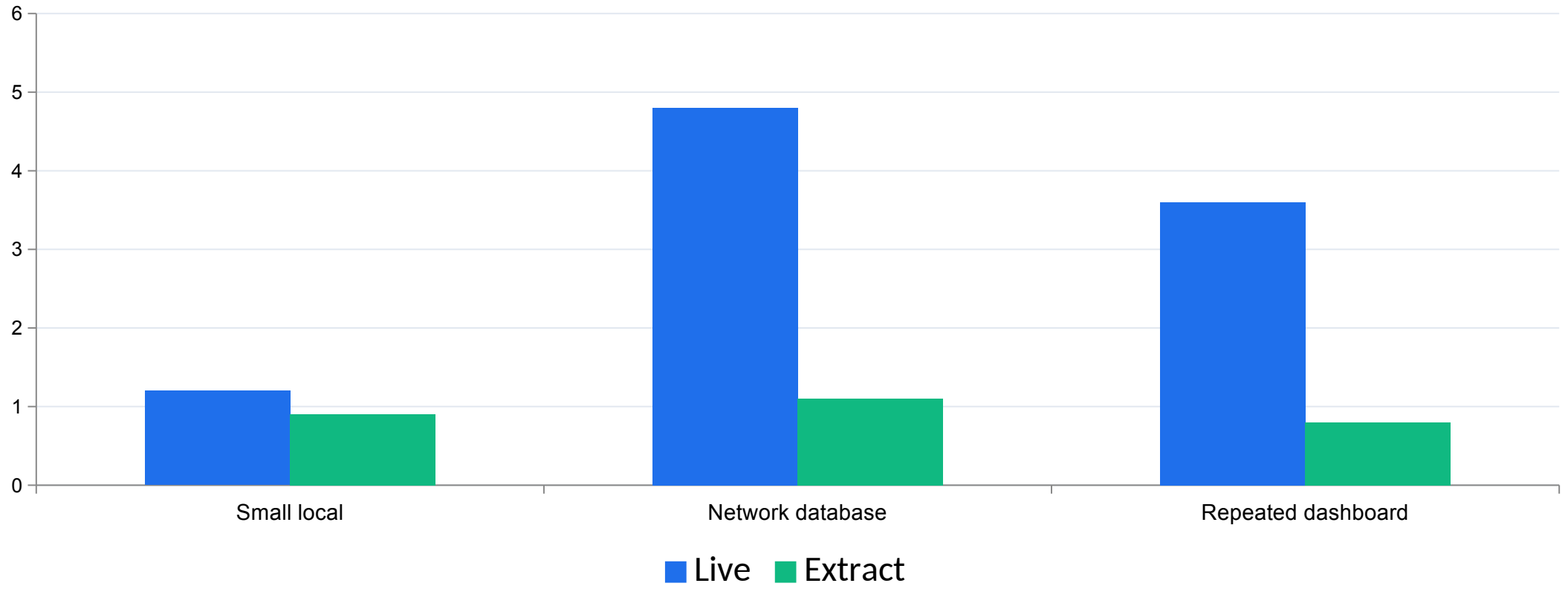
Topic 01 Blueprint

Objective group	Codes	Objectives
Create live connections and extracts	1.1.1–1.1.5	5
Create and manage the data model	1.2.1–1.2.3	4
Manage data properties	1.3.1–1.3.5	5
Cleanse data with Tableau Prep Builder	1.4.1–1.4.10	10

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CONCEPT VISUAL · SYNTHETIC SECONDS

Illustrative Query Response: Live vs Extract



Illustrative only: actual performance depends on source, network, filters, extract design and workload.

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Create live connections and extracts

1

Live

Queries the source as the view runs; current but dependent on source performance and availability.

2

Extract

A local .hyper snapshot optimised for analytics, portability and offline work.

3

.TDS metadata

Stores connection and field metadata; .TDSX can package local dependencies.

4

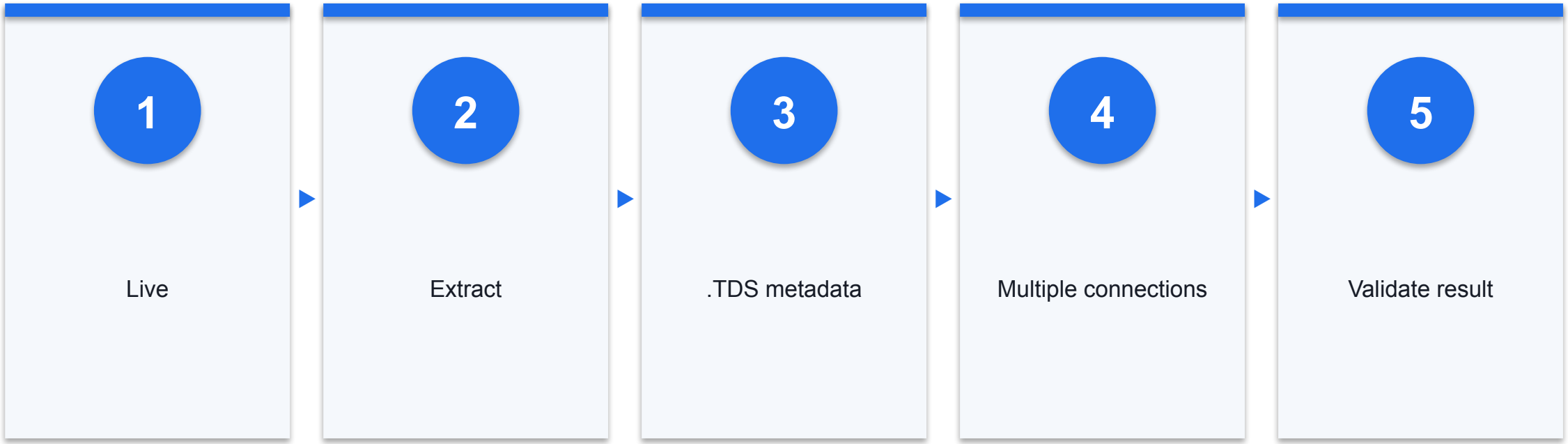
Multiple connections

One Tableau data source can contain several connections and logical tables.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Create live connections and extracts



This is a reasoning map, not a click-by-click lab procedure.

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1.1.1 · Create a live connection

1

What it means

A live connection sends queries to the source during interaction.

2

Use it when

Use when source freshness and central governance matter.

3

Exam clue

Current or real-time scenarios usually point to live.

4

Common pitfall

Live is not automatically faster; network and query design still govern response time.

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1.1.2 · Live versus extract

1

What it means

The choice balances freshness, performance, portability and supported features.

2

Use it when

Use an extract for offline analysis, acceleration or a stable reporting snapshot.

3

Exam clue

Select the mode that serves the scenario's priority.

4

Common pitfall

An extract becomes stale until refreshed; a live connection may still cache results.

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1.1.3 · Create and refresh an extract

1

What it means

An extract materialises selected data in Tableau's columnar .hyper format.

2

Use it when

Filter rows or columns when the complete source is unnecessary.

3

Exam clue

Full refresh rebuilds; incremental refresh adds rows using a reliable key.

4

Common pitfall

Incremental refresh does not update changed historical rows by itself.

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1.1.4 · Save metadata in a .TDS

1

What it means

A Tableau Data Source file stores connection details and field metadata.

2

Use it when

Use it to standardise names, aliases, roles, folders and calculations.

3

Exam clue

TDS contains metadata; TDSX packages metadata with local files or extracts.

4

Common pitfall

Do not confuse a data source definition with a packaged workbook (.twbx).

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1.1.5 · Use multiple connections

1

What it means

A data source may combine file, database or published connections.

2

Use it when

Use when required tables do not reside in the same connection.

3

Exam clue

Connection is transport; data source is the analytical model exposed to sheets.

4

Common pitfall

Cross-database joins can add query cost and connector limitations.

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Create and manage the data model

1

Relationships

Logical links evaluated at query time; preserve each table's level of detail.

2

Joins

Physical row combination using matching clauses and a fixed join type.

3

Unions

Append rows from similarly structured tables; align fields by name and type.

4

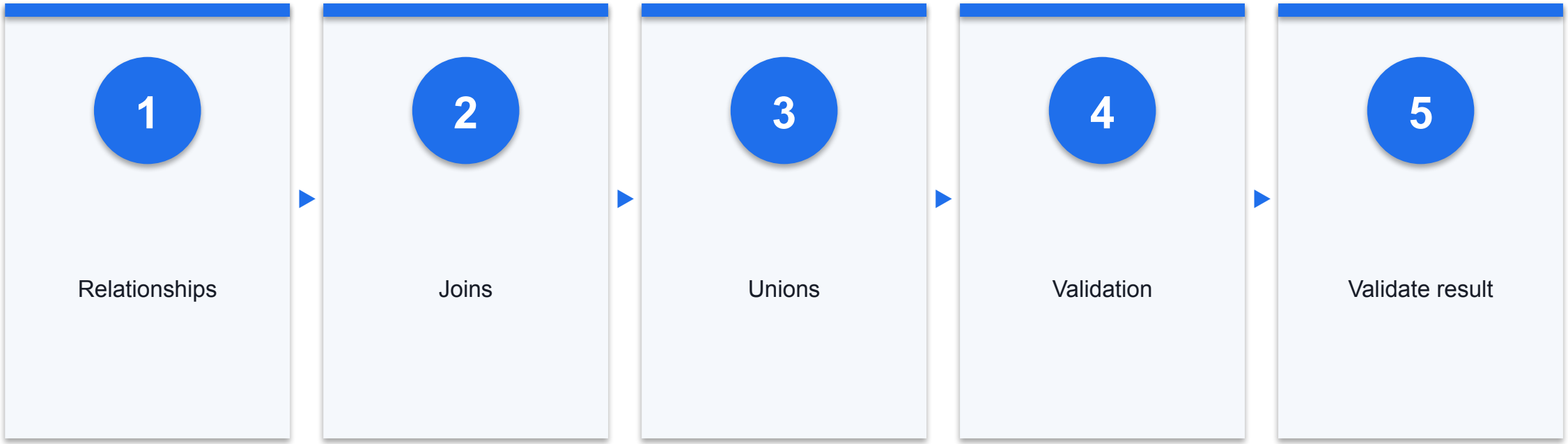
Validation

Compare expected grain, row counts, unmatched keys and nulls before analysis.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Create and manage the data model



This is a reasoning map, not a click-by-click lab procedure.

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1.2.1 · Add relationships

1

What it means

Relationships describe how logical tables correspond without flattening first.

2

Use it when

Use for multi-fact models or tables at different grains.

3

Exam clue

Tableau chooses context-aware joins and aggregation from fields in the view.

4

Common pitfall

Incorrect cardinality assumptions can produce unexpected results.

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1.2.2a · Add joins

1

What it means

A join combines physical tables using matching clauses.

2

Use it when

Use when row-level control and one flattened table are required.

3

Exam clue

Inner keeps matches; left preserves all left rows; right/full follow preservation logic.

4

Common pitfall

Many-to-many joins can multiply rows and inflate measures.

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1.2.2b · Add unions

1

What it means

A union stacks records vertically from compatible tables.

2

Use it when

Use for monthly or regional files with the same columns.

3

Exam clue

Wildcard unions can include similarly named tables or files.

4

Common pitfall

Mismatched names create separate columns until they are merged.

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1.2.3 · Choose join or relationship

1

What it means

Relationships preserve independent grains; joins establish one physical grain.

2

Use it when

Prefer relationships for flexible analysis; use joins for deliberate shaping.

3

Exam clue

Different grains or duplicated measures usually favour relationships.

4

Common pitfall

A relationship line does not guarantee correct keys; validate it.

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Manage data properties

1

Semantic clarity

Rename columns and apply aliases so consumers understand the data.

2

Correct type

Date, number, string and Boolean types determine available operations.

3

Geographic meaning

Roles such as Country and Postal Code enable generated coordinates.

4

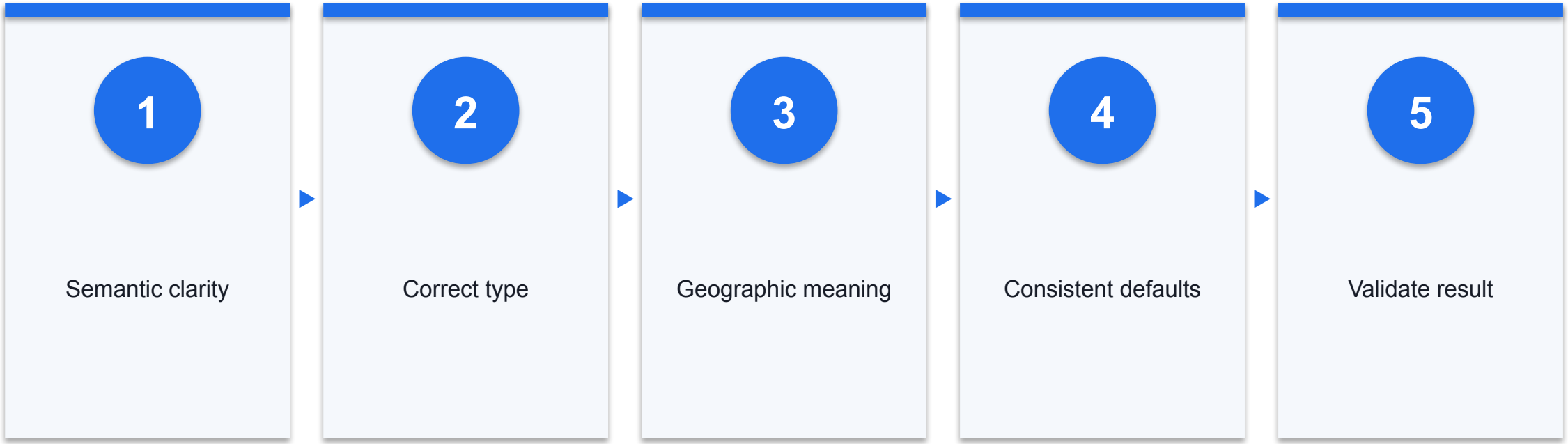
Consistent defaults

Set aggregation, format, colour and date behaviour once for reuse.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Manage data properties



This is a reasoning map, not a click-by-click lab procedure.

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1.3.1 · Rename a field

1

What it means

A caption can be business-friendly without changing the source column.

2

Use it when

Replace codes or database naming conventions with clear language.

3

Exam clue

Renaming affects Tableau metadata, not the source system.

4

Common pitfall

Do not create ambiguous duplicate captions.

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1.3.2 · Assign aliases

1

What it means

Aliases replace displayed member labels while retaining original values.

2

Use it when

Use for abbreviations or coded categories.

3

Exam clue

Aliases apply to discrete dimension members.

4

Common pitfall

Aliases do not cleanse the source value.

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1.3.3 · Assign geographic roles

1

What it means

A geographic role tells Tableau how to geocode a field.

2

Use it when

Use for country, state, city, postal code or custom location analysis.

3

Exam clue

Generated Latitude and Longitude appear after a valid role.

4

Common pitfall

Ambiguous places need country or state context.

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1.3.4 · Change a data type

1

What it means

The type defines valid operations, parsing and visual encodings.

2

Use it when

Correct imported IDs, dates, decimals and Boolean indicators.

3

Exam clue

A calendar-looking string is not a date until parsed correctly.

4

Common pitfall

Conversion can create nulls where values are invalid.

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1.3.5 · Set default properties

1

What it means

Defaults control aggregation, number/date format, colour and sort.

2

Use it when

Use when a field should behave consistently on new sheets.

3

Exam clue

Default aggregation can still be changed in a view.

4

Common pitfall

Currency formatting does not convert currencies or units.

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Cleanse data with Tableau Prep Builder

1

Flow, not clicks

A Prep flow is a repeatable, auditable pipeline of Input, Clean, Pivot, Union, Join, Aggregate and Output steps.

2

Profile before you fix

The profile pane exposes dirty types, outliers, nulls, duplicates and inconsistent members before any change.

3

Change list is the audit trail

Every cleaning action is recorded, reorderable and removable, so the flow stays reproducible.

4

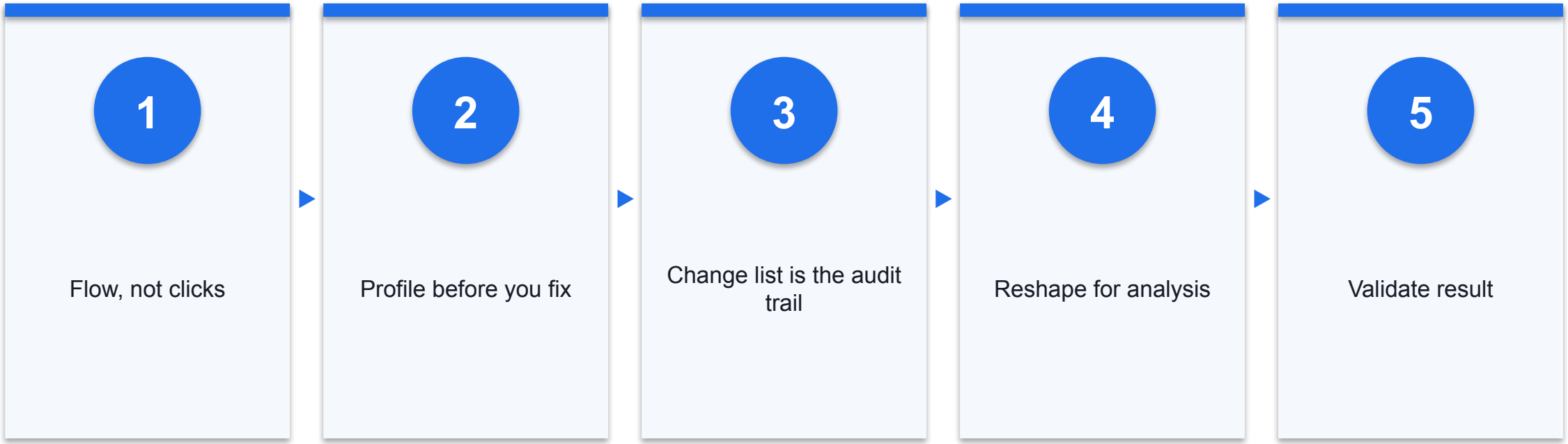
Reshape for analysis

Pivot columns to rows for tidy time series; union files and join lookups to build one analysis-ready table.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Cleanse data with Tableau Prep Builder



This is a reasoning map, not a click-by-click lab procedure.

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1.4.1 · Connect a Prep input and use the Data Interpreter

1

What it means

Input steps bring in Excel, CSV, database and published sources; the Data Interpreter strips titles, notes and blank banding.

2

Use it when

Use when spreadsheets carry cosmetic headers or merged title rows above the real table.

3

Exam clue

The input step is where sampling, field removal and input-level filters are set.

4

Common pitfall

Interpreting an already-clean sheet can drop legitimate header rows; always review the result.

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1.4.2 · Add a Clean step and read the profile pane

1

What it means

The Clean step profiles each field, showing value distribution, nulls and outliers.

2

Use it when

Use it first to see what is actually wrong before deciding on fixes.

3

Exam clue

Clicking a profile bar filters the data grid to those records.

4

Common pitfall

Judging quality from the data grid alone hides the distribution problems the profile reveals.

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1.4.3 · Fix data types and repair non-numeric text

1

What it means

Currency, thousands separators and stray characters block numeric conversion.

2

Use it when

Use a calculated field such as a REPLACE to strip commas before changing the type.

3

Exam clue

A field that refuses to become a number usually contains punctuation or spaces.

4

Common pitfall

Converting directly to a number silently produces nulls where parsing fails.

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1.4.4 · Apply cleaning operations

1

What it means

Prep offers Remove Letters, Remove Numbers, Remove Punctuation, Trim Spaces, Remove Extra Spaces and case conversion.

2

Use it when

Use to normalise names, codes and free-text categories.

3

Exam clue

Remove Extra Spaces collapses repeats; Remove All Spaces deletes every space.

4

Common pitfall

Removing all spaces destroys multi-word names that still need to be readable.

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1.4.5 · Split fields automatically and manually

1

What it means

Automatic split infers a separator; custom split specifies the delimiter and number of columns.

2

Use it when

Use to separate names, codes, dates or concatenated attributes.

3

Exam clue

Split creates new fields; the original is usually removed afterwards.

4

Common pitfall

An inferred split breaks when the separator is inconsistent across rows.

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1.4.6 · Filter and handle nulls and duplicates

1

What it means

Filters can be by value, wildcard, range, calculation or null status; duplicates are found by profiling the row key.

2

Use it when

Use to exclude test rows, negative sales, out-of-scope regions and empty keys.

3

Exam clue

Keep Only and Exclude are the fastest filters directly from the profile.

4

Common pitfall

Filtering away nulls can silently drop valid records that merely lack one attribute.

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1.4.7 · Group and Replace to standardise members

1

What it means

Group and Replace merges variants using manual selection, pronunciation, common characters or spelling similarity.

2

Use it when

Use for misspelt city, product or customer names that should be one member.

3

Exam clue

Fuzzy matching proposes groups; the analyst confirms them.

4

Common pitfall

Aggressive fuzzy matching can merge genuinely different members.

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1.4.8 · Pivot columns to rows

1

What it means

A Pivot step converts wide month or category columns into a tidy field-and-value pair.

2

Use it when

Use when each period sits in its own column instead of its own row.

3

Exam clue

Pivoted output creates one row per original column value.

4

Common pitfall

Leaving data wide forces one calculation per column and breaks date logic.

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OFFICIAL OBJECTIVE · DOMAIN 1

1.4.9 · Union, join and aggregate in a flow

1

What it means

Union appends compatible files; Join matches on keys; Aggregate rolls rows to a coarser grain.

2

Use it when

Use to combine monthly or country files and attach lookup attributes.

3

Exam clue

Mismatched field names appear in the union as separate columns until merged.

4

Common pitfall

A join in Prep multiplies rows the same way a join in Desktop does.

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1.4.10 · Create calculated fields and output the flow

1

What it means

Prep calculations derive new columns; Output writes a CSV, Excel, .hyper extract or published data source.

2

Use it when

Use to compute derived measures and publish a governed, refreshable clean table.

3

Exam clue

Running the flow re-executes every step in order against current input.

4

Common pitfall

Outputting without re-running the flow ships stale results.

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LAB 01 · TOPIC 01 · OVERVIEW ONLY

Audit a Retail Sales Connection and Extract

Northstar Retail needs a trustworthy weekly sales source before analysts build regional reporting.

1. **Connect workbook**
2. **Audit types**
3. **Set roles**
4. **Create extract**
5. **Validate refresh**

OUTPUT A validated Tableau data source, field-role audit, live/extract decision and refreshed .hyper extract

Detailed click-by-click procedure: Learner Guide and this lab folder · Tools: **Tableau Desktop · Excel · live connection · extract · metadata**

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VERIFY THE OUTCOME

Lab 01 — Acceptance Test

✓ Evidence standard

The source has the expected row count and date range, geographic fields are recognised, currency/percentage defaults are correct, and the extract refreshes without errors.

Submit the Tableau workbook, requested screenshots/exports and completed acceptance-checklist.md from the lab folder.

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LAB 02 · TOPIC 01 · OVERVIEW ONLY

Model Orders, Returns, Products and Targets

GreenBasket Grocers wants sales, returns and monthly targets analysed without duplicating measures across different table grains.

1. Inspect grains	2. Relate tables	3. Validate keys	4. Compare join	5. Prove totals

OUTPUT A four-table logical model with validated cardinality, unmatched-key checks and a join-risk comparison

Detailed click-by-click procedure: Learner Guide and this lab folder · Tools: **Tableau Desktop · relationships · joins · referential integrity · grain validation**

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VERIFY THE OUTCOME

Lab 02 — Acceptance Test

✓ Evidence standard

Sales matches the control total, Return Count matches the Returns table, targets do not duplicate, and the learner can explain why relationships are safer for the mixed-grain model.

Submit the Tableau workbook, requested screenshots/exports and completed acceptance-checklist.md from the lab folder.

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LAB 03 · TOPIC 01 · OVERVIEW ONLY

Union Monthly Files and Join Store Attributes

MetroMart receives separate monthly transaction tabs plus a store master and needs one analysis-ready source.

1. Union months
2. Inspect mismatch
3. Merge fields
4. Join stores
5. Save metadata

OUTPUT A three-month union with provenance fields, merged columns, a validated store join and saved data-source metadata

Detailed click-by-click procedure: Learner Guide and this lab folder · Tools: **Tableau Desktop · unions · wildcard patterns · joins · aliases · .tds/.tdsx**

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VERIFY THE OUTCOME

Lab 03 — Acceptance Test

✓ Evidence standard

All three months appear, merged sales has no avoidable nulls, record counts match Control, the left join preserves transactions, and the saved source opens with metadata intact.

Submit the Tableau workbook, requested screenshots/exports and completed acceptance-checklist.md from the lab folder.

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LAB 11 · TOPIC 01 · OVERVIEW ONLY

Cleanse a Messy Sales Extract with Tableau Prep Builder

Harbourfront Trading exports raw transactions with currency text, inconsistent city spellings, split name fields, duplicate rows and wide monthly budget columns. Analysts need one governed, refreshable clean table before any Tableau Desktop analysis.

1. Profile input	2. Fix types	3. Clean and split	4. Group members	5. Pivot and join	6. Output flow

OUTPUT A saved Tableau Prep flow (.tfl) with Input, Clean, Split, Group and Replace, Pivot, Union, Join and Output steps, plus a cleaned .hyper or CSV output and a documented change list

Detailed click-by-click procedure: Learner Guide and this lab folder · Tools: **Tableau Prep Builder · profile pane · clean step · split · group and replace · pivot · union · join · output**

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VERIFY THE OUTCOME

Lab 11 — Acceptance Test

✓ Evidence standard

The flow runs end to end without errors; Price Paid is numeric, duplicates and invalid rows are removed, city members are standardised, budgets are pivoted long, the union has no avoidable nulls, the join does not multiply rows, and the output reconciles to the Control sheet.

Submit the Tableau workbook, requested screenshots/exports and completed acceptance-checklist.md from the lab folder.

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RECAP

Topic 01 — Exam Check

1

Explain

Define the key ideas in connecting to and preparing data in tableau desktop.

2

Choose

Select the right feature from a short business scenario.

3

Predict

Reason about the result of shelf, model or formatting changes.

4

Prove

Use the lab workbook and acceptance evidence to validate the outcome.

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02

TOPIC 02

Exploring and Analyzing Data in Tableau Desktop

Official exam weighting 37% · Charts · groups · sets · hierarchies · filters · analytics · calculations

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EXAM WEIGHTING 37%

Topic 02 — Core Ideas

1

Question before chart

Bars compare, lines show time, scatterplots show relationships and maps show location.

2

Structure narrows meaning

Groups, sets, hierarchies, sorts and filters determine the population compared.

3

Analytics adds context

Lines, totals, bins, calculations and parameters turn marks into evidence.

4

Order matters

Context, dimension, measure and table-calculation filters execute at different stages.

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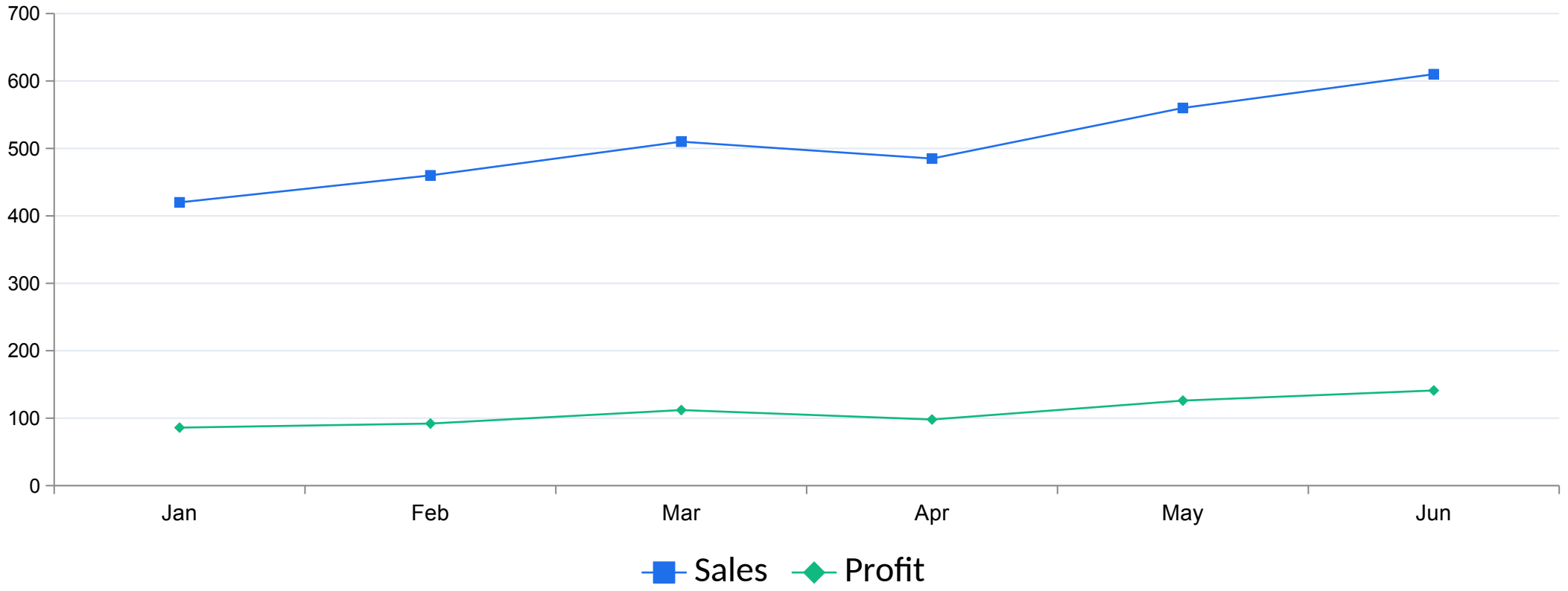
Topic 02 Blueprint

Objective group	Codes	Objectives
Create basic charts	2.1.1–2.1.9	9
Organise data and apply filters	2.2.1–2.2.5	5
Apply analytics to a worksheet	2.3.1–2.3.7	7

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EDITABLE POWERPOINT CHART · LAB PATTERN

Synthetic Monthly Sales and Profit



Synthetic SGD thousands — the chart demonstrates trend selection, not a real company result.

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Create basic charts

1

Comparison

Bars and crosstabs expose magnitude and exact values.

2

Trend

Lines show change across ordered time; continuous dates create axes.

3

Relationship

Scatterplots reveal association, clusters and outliers.

4

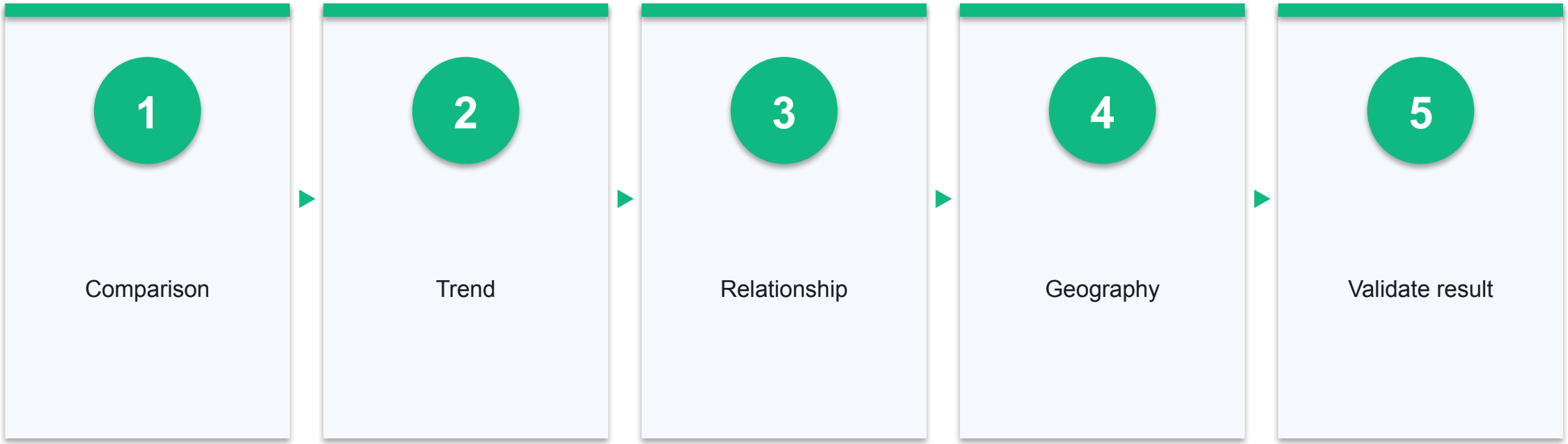
Geography

Symbol, filled and density maps answer different spatial questions.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Create basic charts



This is a reasoning map, not a click-by-click lab procedure.

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2.1.1 · Create a bar chart

1

What it means

Bars encode magnitude by length along a common baseline.

2

Use it when

Use for ranking and comparing categories.

3

Exam clue

One dimension plus one measure is the canonical pattern.

4

Common pitfall

Avoid 3-D effects, too many categories and unsorted ranking views.

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2.1.2 · Create a line chart

1

What it means

Lines connect ordered points to emphasise change over time.

2

Use it when

Use for trends, seasonality and rate of change.

3

Exam clue

Continuous dates form axes; discrete date parts form headers.

4

Common pitfall

Do not connect unrelated categories as a sequence.

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2.1.3 · Create a scatterplot

1

What it means

Two quantitative axes show the relationship between measures.

2

Use it when

Use to assess correlation, clusters and outliers.

3

Exam clue

Add a dimension to Detail for one mark per entity.

4

Common pitfall

Aggregation may collapse records into one mark.

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2.1.4 · Create a map

1

What it means

Geographic fields generate spatial marks using Tableau geocoding.

2

Use it when

Use when location contributes to the question.

3

Exam clue

Symbols compare points; fills compare areas; density shows concentration.

4

Common pitfall

A bar chart is more accurate when geography adds no meaning.

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2.1.5 · Create a combined-axis chart

1

What it means

Multiple measures share one axis and Marks card.

2

Use it when

Use when compatible units should be compared directly.

3

Exam clue

Measure Values and Measure Names are common building blocks.

4

Common pitfall

Different units on one axis mislead.

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2.1.6 · Create a dual-axis chart

1

What it means

Two measures use independent axes with separate Marks cards.

2

Use it when

Use to layer related measures with different scales or mark types.

3

Exam clue

Synchronise axes when measures share units.

4

Common pitfall

Unsynchronised axes can imply artificial relationships.

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2.1.7 · Create a stacked bar

1

What it means

Segments divide each bar into contributions to a total.

2

Use it when

Use for composition plus total comparison.

3

Exam clue

Colour defines the stack.

4

Common pitfall

Middle segments lack a common baseline.

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2.1.8 · Create a density map

1

What it means

Overlapping points become a smooth concentration surface.

2

Use it when

Use where hotspots matter more than individual points.

3

Exam clue

Intensity reflects local concentration and optional weighting.

4

Common pitfall

Sparse data or smoothing can manufacture hotspots.

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2.1.9 · Crosstab and highlight table

1

What it means

Text tables show exact values; colour adds pattern detection.

2

Use it when

Use for lookup precision plus comparative scanning.

3

Exam clue

Measure Names and Measure Values can present several KPIs.

4

Common pitfall

Too many cells overwhelm; filter and use meaningful scales.

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Organise data and apply filters

1

Groups

Combine members into a new categorical field.

2

Sets

Define an IN/OUT subset that can be dynamic or combined.

3

Hierarchies

Provide a drill path from broad to detailed dimensions.

4

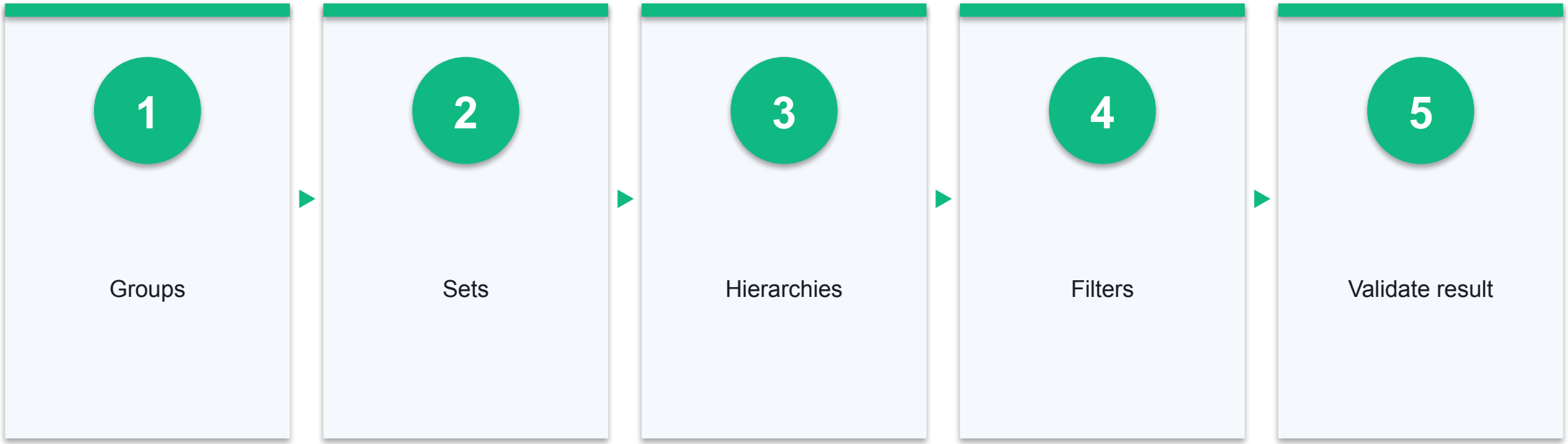
Filters

Control which records or aggregated marks remain.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Organise data and apply filters



This is a reasoning map, not a click-by-click lab procedure.

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2.2.1 · Create groups

1

What it means

Groups combine members into higher-level categories.

2

Use it when

Use to correct labels, merge small categories or define business groupings.

3

Exam clue

Create from marks, headers or the Data pane.

4

Common pitfall

A manual group does not absorb new members automatically.

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2.2.2 · Create sets

1

What it means

A set divides members into IN and OUT using rules or selection.

2

Use it when

Use for Top N, threshold cohorts and set actions.

3

Exam clue

Sets can be fixed, conditional, Top N or combined.

4

Common pitfall

A set is not just a filter; it can remain visible as a category.

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2.2.3 · Create hierarchies

1

What it means

Hierarchies organise dimensions into a drill path.

2

Use it when

Use Country > Region > City or Category > Product.

3

Exam clue

Plus/minus controls expand or collapse levels.

4

Common pitfall

A hierarchy does not define data-model integrity.

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2.2.4 · Add filters

1

What it means

Filters restrict data or marks at specific order-of-operations stages.

2

Use it when

Use to focus scope or support interaction.

3

Exam clue

Dimension, measure, context and table-calculation filters differ.

4

Common pitfall

Filtering after aggregation is not filtering source rows.

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2.2.5 · Add date filters

1

What it means

Dates can be filtered by part, range, relative period or starting date.

2

Use it when

Use for rolling windows and time scope controls.

3

Exam clue

Relative date filters move with the current date.

4

Common pitfall

Discrete Month alone mixes the same month across years.

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Apply analytics to a worksheet

1. **Sort and rank**
   Make comparisons intentional and surface priority.

2. **Reference and totals**
   Add baselines, bands, subtotals and grand totals.

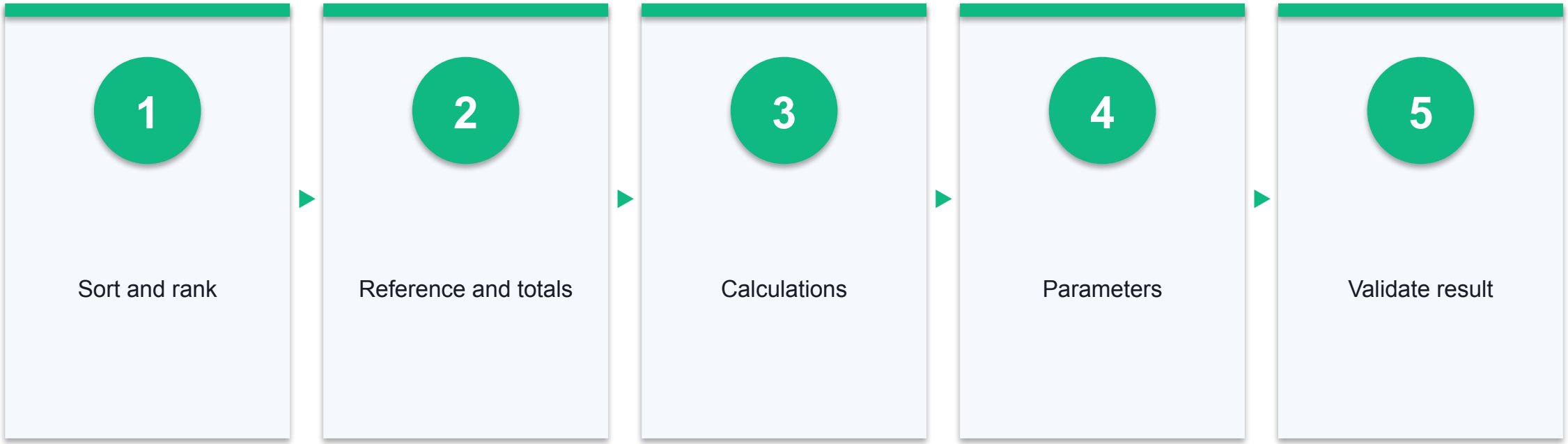
3. **Calculations**
   Use row-level, aggregate and table logic at the right grain.

4. **Parameters**
   Let one value steer calculations, filters or reference lines.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Apply analytics to a worksheet



This is a reasoning map, not a click-by-click lab procedure.

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2.3.1 · Manual and computed sorts

1

What it means

Sorting controls the order of headers and marks.

2

Use it when

Use manual order for business logic and computed order for ranking.

3

Exam clue

Nested sorts order members within panes.

4

Common pitfall

Manual order can become stale as data changes.

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2.3.2 · Add a reference line

1

What it means

A line or band provides a benchmark for a scope.

2

Use it when

Use for targets, averages, medians and ranges.

3

Exam clue

Scope can be table, pane or cell.

4

Common pitfall

A table average can hide pane-level differences.

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2.3.3 · Use quick table calculations

1

What it means

Table calculations operate on marks already returned to the view.

2

Use it when

Use running total, percent of total, difference, rank or moving average.

3

Exam clue

Compute Using defines addressing and partitioning.

4

Common pitfall

Changing layout can change the calculation.

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2.3.4 · Create bins and histograms

1

What it means

Bins partition a measure into equal-width intervals.

2

Use it when

Use a histogram for distribution shape and unusual ranges.

3

Exam clue

Bin size controls resolution.

4

Common pitfall

Poor bin width can hide or invent patterns.

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2.3.5 · Create calculated fields

1

What it means

Calculated fields derive row-level, aggregate or table values.

2

Use it when

Use for arithmetic, strings, dates and business rules.

3

Exam clue

Aggregate and non-aggregate expressions cannot be mixed directly.

4

Common pitfall

Nulls propagate unless handled.

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2.3.6 · Use parameters

1

What it means

A parameter is a workbook value controlled by the user.

2

Use it when

Use to swap measures, set thresholds or run what-if analysis.

3

Exam clue

A parameter acts only when referenced.

4

Common pitfall

Parameters are not automatically row filters.

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2.3.7 · Display totals

1

What it means

Subtotals and grand totals summarise selected levels.

2

Use it when

Use when audiences need components and overall results.

3

Exam clue

Total All Using can change aggregation.

4

Common pitfall

Summed ratios and percentages often create invalid totals.

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LAB 04 · TOPIC 02 · OVERVIEW ONLY

Build the Core Tableau Chart Portfolio

A consumer-electronics team needs views for ranking, trend, relationship, geography, composition and precise lookup.

1. Frame question

2. Choose chart

3. Build marks

4. Format honestly

5. Explain insight

OUTPUT Nine named worksheets: bar, line, scatterplot, symbol map, combined axis, dual axis, stacked bar, density map and highlight table

Detailed click-by-click procedure: Learner Guide and this lab folder · Tools: **Tableau Desktop · Show Me · Rows/Columns · Marks · Measure Names/Values**

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VERIFY THE OUTCOME

Lab 04 — Acceptance Test

✓ Evidence standard

All nine chart types are present, each matches its stated question, axes and units are honest, maps resolve locations, and each view has a concise evidence-based insight.

Submit the Tableau workbook, requested screenshots/exports and completed acceptance-checklist.md from the lab folder.

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LAB 05 · TOPIC 02 · OVERVIEW ONLY

Create Groups, Sets, Hierarchies and Filters

HarbourLife Insurance needs a reusable segmentation view for advisers and portfolio managers.

1. Group tiers	2. Define sets	3. Build hierarchy	4. Scope filters	5. Test order

OUTPUT Customer tier groups, Top-N and threshold sets, drill hierarchy, context filter and relative date controls

Detailed click-by-click procedure: Learner Guide and this lab folder · Tools: **Tableau Desktop · groups · sets · hierarchy · context · date filters**

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VERIFY THE OUTCOME

Lab 05 — Acceptance Test

✓ Evidence standard

The two hierarchies drill correctly, sets produce expected membership, relative dates use the reporting date, and the learner can explain how context changes Top-N evaluation.

Submit the Tableau workbook, requested screenshots/exports and completed acceptance-checklist.md from the lab folder.

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LAB 06 · TOPIC 02 · OVERVIEW ONLY

Apply Calculations, Table Calculations and Parameters

Crescent Hospitality wants profitability, occupancy and performance trends with user-controlled thresholds.

1. Define grain	2. Write calc	3. Set compute	4. Add parameter	5. Validate totals

OUTPUT Calculated KPIs, running totals, percent-of-total, moving average, reference line and parameter-driven measure swap

Detailed click-by-click procedure: Learner Guide and this lab folder · Tools: **Tableau Desktop · calculated fields · table calculations · parameters · reference lines · totals**

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VERIFY THE OUTCOME

Lab 06 — Acceptance Test

✓ Evidence standard

Calculations match the Control sheet, table calculations keep intended addressing/partitioning, parameters affect the view, and totals use mathematically valid logic.

Submit the Tableau workbook, requested screenshots/exports and completed acceptance-checklist.md from the lab folder.

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LAB 07 · TOPIC 02 · OVERVIEW ONLY

Analyse Southeast Asia Logistics Geographically

Meridian Logistics wants to compare shipment volume, delay rates and port congestion across Southeast Asia.

1. **Resolve places**
2. **Map points**
3. **Encode metrics**
4. **Show density**
5. **Challenge map**

OUTPUT Resolved geographic roles, three map types, tooltip context and a map-vs-bar decision note

Detailed click-by-click procedure: Learner Guide and this lab folder · Tools: **Tableau Desktop · geographic roles · generated coordinates · density · tooltips**

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VERIFY THE OUTCOME

Lab 07 — Acceptance Test

✓ Evidence standard

Locations resolve correctly, map types match the questions, tooltips carry operational context, colour scales are meaningful, and the learner can state when the bar chart is preferable.

Submit the Tableau workbook, requested screenshots/exports and completed acceptance-checklist.md from the lab folder.

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RECAP

Topic 02 — Exam Check

1

Explain

Define the key ideas in exploring and analyzing data in tableau desktop.

2

Choose

Select the right feature from a short business scenario.

3

Predict

Reason about the result of shelf, model or formatting changes.

4

Prove

Use the lab workbook and acceptance evidence to validate the outcome.

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03

TOPIC 03

Sharing Insights with Dashboards and Workbooks

Official exam weighting 25% · Formatting · dashboards · actions · device layouts · stories · export · publish

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EXAM WEIGHTING 25%

Topic 03 — Core Ideas

1

Format for meaning

Colour, type, shape, size and animation communicate data, not decoration.

2

Design interaction

Filters, highlighters, tooltips and actions answer predictable questions.

3

Layout is a system

Containers, hierarchy and device layouts govern alignment and reading order.

4

Share with intent

Package, publish or export while protecting sensitive detail.

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OFFICIAL SCOPE

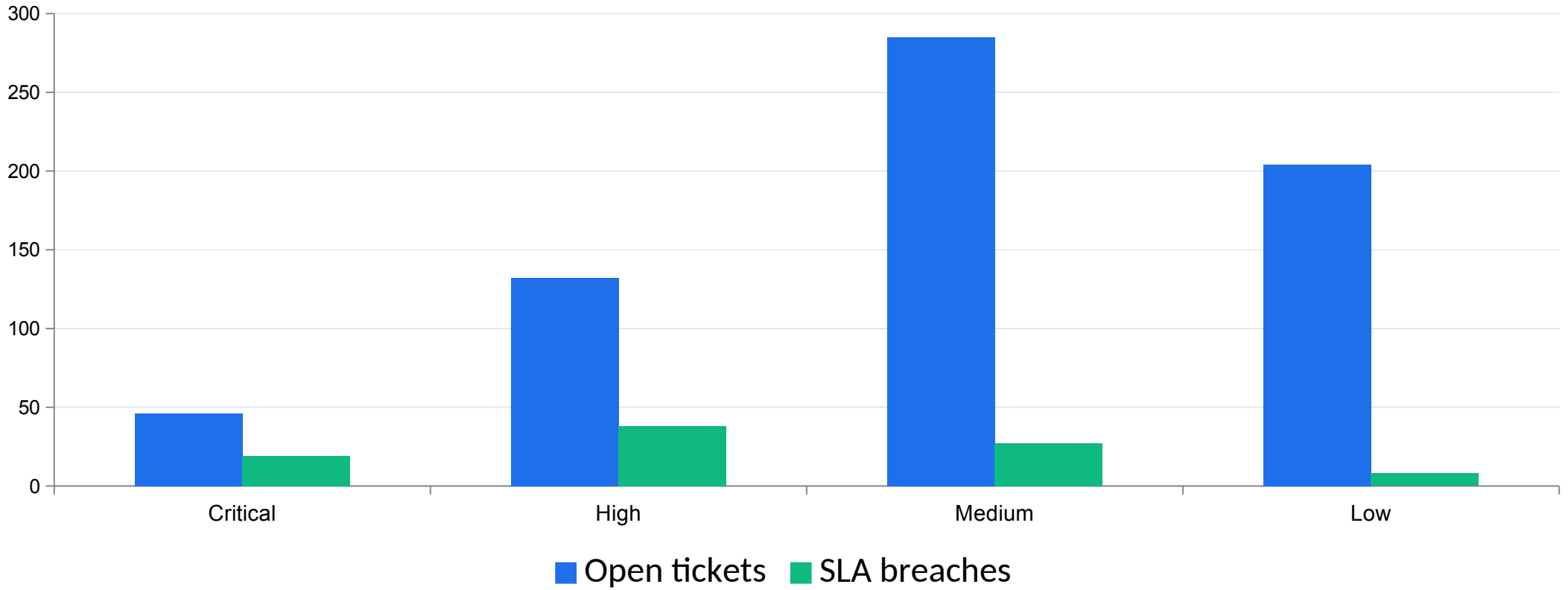
Topic 03 Blueprint

Objective group	Codes	Objectives
Format views for presentation	3.1.1–3.1.6	6
Create and modify a dashboard	3.2.1–3.2.5	5
View and share workbook data	3.3.1–3.3.3	3

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EDITABLE POWERPOINT CHART · SYNTHETIC DATA

Dashboard Attention Should Follow Risk



A dashboard should prioritise decisions, not give equal visual weight to every metric.

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Format views for presentation

1

Colour

Categorical hues show identity; ordered palettes show magnitude.

2

Typography

Create a readable hierarchy and concise labels.

3

Marks

Shape and size encode defined variables and remain distinguishable.

4

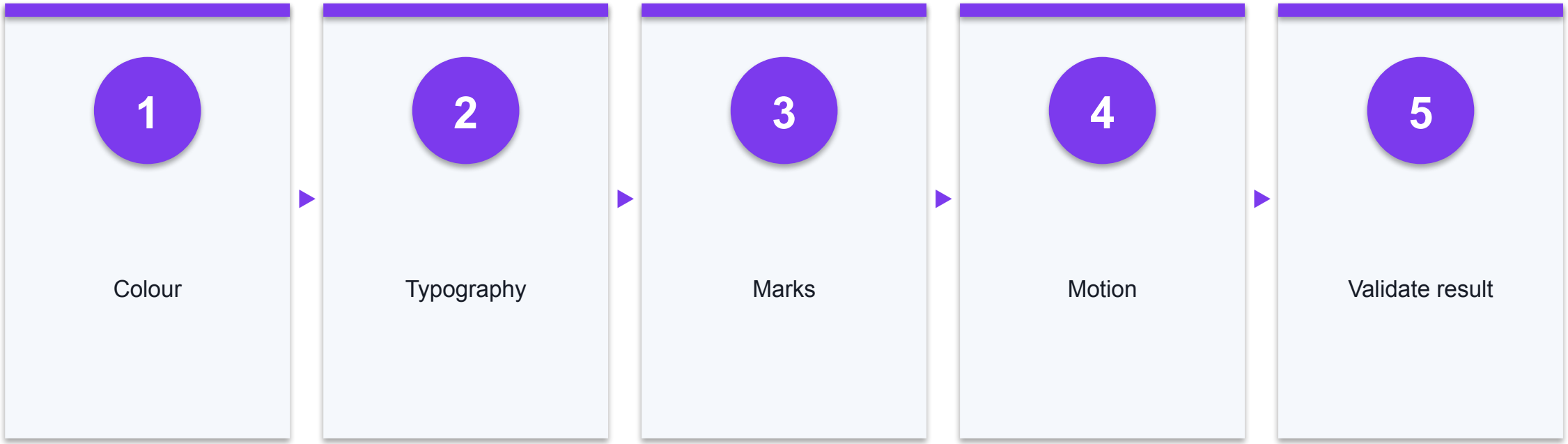
Motion

Animation explains change but must not conceal the final state.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Format views for presentation



This is a reasoning map, not a click-by-click lab procedure.

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3.1.1 · Use colour

1

What it means

Colour can encode category, magnitude or emphasis.

2

Use it when

Use hues for categories and ordered palettes for measures.

3

Exam clue

Legend type reveals discrete or continuous encoding.

4

Common pitfall

Rainbow palettes and low contrast impair interpretation.

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3.1.2 · Configure fonts

1

What it means

Typography establishes hierarchy across titles, labels and tooltips.

2

Use it when

Use consistent families and display-appropriate sizes.

3

Exam clue

Formatting applies at workbook, sheet, pane or element scope.

4

Common pitfall

Tiny labels and excessive bolding reduce scanability.

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3.1.3 · Format marks as shapes

1

What it means

Shape is a categorical channel for point marks.

2

Use it when

Use for a small number of distinguishable categories.

3

Exam clue

Custom shapes can be loaded and assigned.

4

Common pitfall

Shape alone is weak for order and accessibility.

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3.1.4 · Configure viz animations

1

What it means

Animations interpolate filter, sort and data changes.

2

Use it when

Use to help audiences track moving marks.

3

Exam clue

Workbook formatting controls duration and style.

4

Common pitfall

Motion should not be required for understanding.

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3.1.5 · Change mark size

1

What it means

Size can encode magnitude or improve legibility.

2

Use it when

Use measure-driven size for proportional symbols.

3

Exam clue

Area, not radius, is perceived in circles.

4

Common pitfall

Large marks overlap and small marks disappear.

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3.1.6 · Show and hide legends

1

What it means

Legends decode visual channels and can highlight marks.

2

Use it when

Show when encoding is not self-evident; hide redundant cards.

3

Exam clue

Use the card menu or Analysis > Legends.

4

Common pitfall

Removing a necessary legend makes marks ambiguous.

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Create and modify a dashboard

1. **Worksheets**
   Place views that support one decision purpose.

2. **Objects**
   Use tiled/floating containers and navigation deliberately.

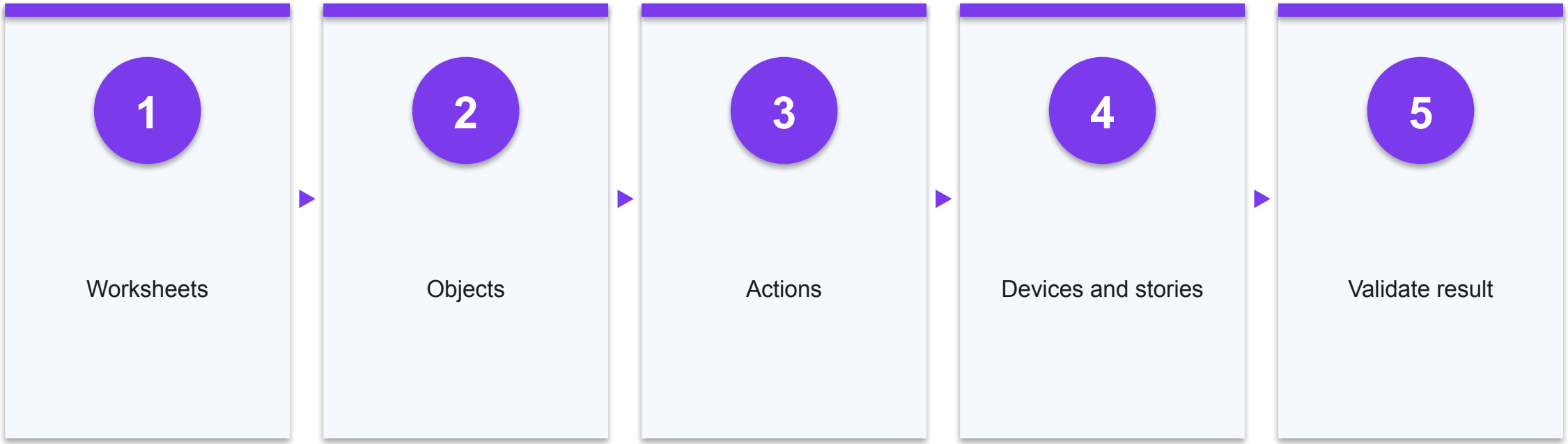
3. **Actions**
   Filter, highlight, parameter, set and URL actions define experience.

4. **Devices and stories**
   Adapt layouts for screens; sequence a guided narrative.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Create and modify a dashboard



This is a reasoning map, not a click-by-click lab procedure.

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3.2.1 · Add worksheets to a dashboard

1

What it means

A dashboard composes sheets and objects into one canvas.

2

Use it when

Create a clear hierarchy with the key view first.

3

Exam clue

Tiled items join containers; floating items use absolute position.

4

Common pitfall

Adding every sheet creates noise.

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3.2.2 · Add consumer interactions

1

What it means

Filters, highlighters, tooltips and navigation support exploration.

2

Use it when

Use controls that match likely questions and correct scope.

3

Exam clue

Apply to Worksheets defines controlled sheets.

4

Common pitfall

Too many controls create contradictory states.

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3.2.3 · Add dashboard actions

1

What it means

Actions respond to select, hover or menu events.

2

Use it when

Use filter, highlight, parameter, set and URL actions.

3

Exam clue

Define source, target, event and clearing behaviour.

4

Common pitfall

Wrong clearing behaviour can leave blank dashboards.

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3.2.4 · Configure layouts and devices

1

What it means

Containers and item hierarchy govern responsive alignment.

2

Use it when

Use fixed/range/automatic sizes and device layouts.

3

Exam clue

Phone layouts can rearrange or omit objects.

4

Common pitfall

Automatic sizing alone does not ensure mobile usability.

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3.2.5 · Create a story

1

What it means

A story sequences sheets or dashboards as annotated points.

2

Use it when

Use for guided explanation rather than open exploration.

3

Exam clue

Each point stores a view state and caption.

4

Common pitfall

Stories do not replace a coherent dashboard.

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View and share workbook data

1

.twb and .twbx

TWB stores XML metadata; TWBX packages local dependencies.

2

Publish

Server/Cloud adds permissions, refresh, governance and browser use.

3

Export

PDF, image and PowerPoint are static communication formats.

4

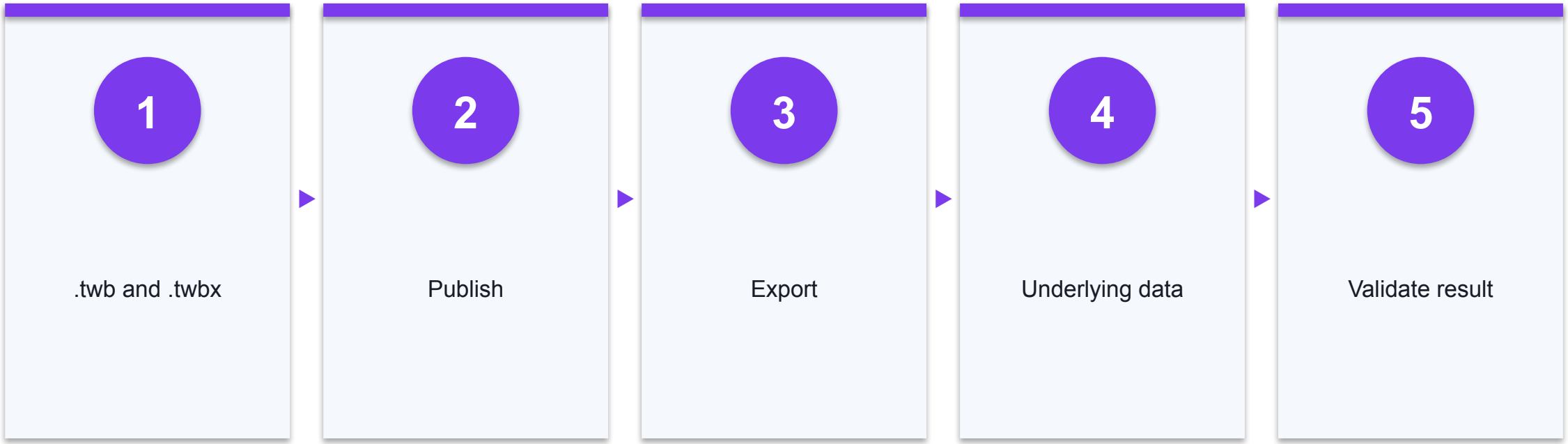
Underlying data

Summary/full-data export can expose detail; verify privacy.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — View and share workbook data



This is a reasoning map, not a click-by-click lab procedure.

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3.3.1 · Share a workbook

1

What it means

Workbooks can be saved, packaged, exported or published.

2

Use it when

Use TWBX for portability and Server/Cloud for governed collaboration.

3

Exam clue

A TWB may break if recipients cannot reach dependencies.

4

Common pitfall

Publishing does not automatically make content public.

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3.3.2 · View and export underlying data

1

What it means

View Data exposes summary and permitted underlying rows.

2

Use it when

Use to audit marks or export support records.

3

Exam clue

Selected marks and permissions define available data.

4

Common pitfall

Exports can leak sensitive detail.

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3.3.3 · Export to PowerPoint

1

What it means

Tableau exports views as static PowerPoint images.

2

Use it when

Use for offline briefings without interactivity.

3

Exam clue

Appearance is preserved, not filters or actions.

4

Common pitfall

Label snapshots with a data-as-of date.

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LAB 08 · TOPIC 03 · OVERVIEW ONLY

Build an Interactive Service Operations Dashboard

ApexCare's service director needs a daily dashboard for backlog, SLA risk, channel demand and team performance.

1. **Frame decision**
2. **Compose views**
3. **Arrange layout**
4. **Add actions**
5. **Test consumers**

OUTPUT A 1366×768 operational dashboard with KPIs, four views, scoped filters, highlight/filter actions and accessible tooltips

Detailed click-by-click procedure: Learner Guide and this lab folder · Tools: **Tableau Desktop · dashboards · layout containers · filters · actions · tooltips**

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VERIFY THE OUTCOME

Lab 08 — Acceptance Test

✓ Evidence standard

The dashboard loads in the target size, KPIs match Control, filters are scoped, actions clear correctly, no object overlaps, and a manager can answer the three stated operational questions.

Submit the Tableau workbook, requested screenshots/exports and completed acceptance-checklist.md from the lab folder.

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LAB 09 · TOPIC 03 · OVERVIEW ONLY

Optimise, Package and Share an Executive Workbook

Sentinel Energy must brief executives while giving analysts a governed interactive workbook.

1. Format view	2. Design devices	3. Build story	4. Package files	5. Govern sharing

OUTPUT Desktop and phone layouts, three story points, TWB/TWBX comparison, PDF/PNG/PPT exports and a publishing checklist

Detailed click-by-click procedure: Learner Guide and this lab folder · Tools: **Tableau Desktop · Device Designer · stories · TWBX · PDF/PNG/PPT · Server/Cloud**

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VERIFY THE OUTCOME

Lab 09 — Acceptance Test

✓ Evidence standard

Desktop and phone layouts are usable, the story has three coherent points, TWBX is portable, static exports are labelled with data-as-of context, and the publish plan protects underlying data.

Submit the Tableau workbook, requested screenshots/exports and completed acceptance-checklist.md from the lab folder.

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RECAP

Topic 03 — Exam Check

1. **Explain**
   Define the key ideas in sharing insights with dashboards and workbooks.

2. **Choose**
   Select the right feature from a short business scenario.

3. **Predict**
   Reason about the result of shelf, model or formatting changes.

4. **Prove**
   Use the lab workbook and acceptance evidence to validate the outcome.

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04

TOPIC 04

Understanding Tableau Concepts and Certification Preparation

Official exam weighting 15% · Dimensions · measures · discrete · continuous · aggregation · exam strategy

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EXAM WEIGHTING 15%

Topic 04 — Core Ideas

1

Role and type differ

Dimension/measure is analytical role; discrete/continuous is display behaviour.

2

Blue and green

Blue discrete pills make headers; green continuous pills make axes.

3

Aggregation follows grain

Measures aggregate at the level defined by dimensions.

4

Reason from consequences

Use pill colour, shelf placement, grain and order of operations.

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OFFICIAL SCOPE

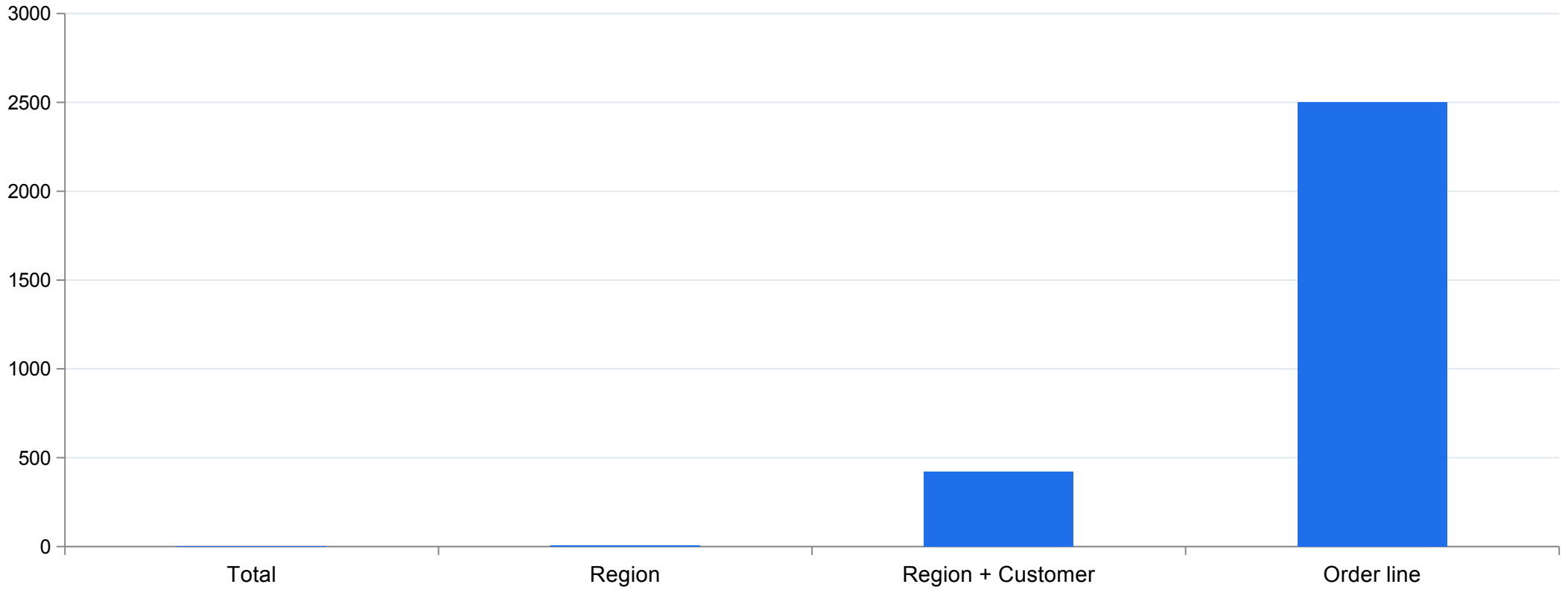
Topic 04 Blueprint

Objective group	Codes	Objectives
Understand dimensions and measures	4.1.1–4.1.3	3
Understand discrete and continuous fields	4.2.1–4.2.3	3
Understand aggregations	4.3.1–4.3.2	2

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EDITABLE POWERPOINT CHART · CONCEPT VISUAL

Dimensions Change the Level of Detail



Adding dimensions partitions records into finer marks; it does not create new source rows.

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Understand dimensions and measures

1

Dimension

Qualitative field that slices, groups or labels marks.

2

Measure

Quantitative field usually aggregated across dimensions.

3

Convertible role

A numeric ID may be a dimension; categories can be counted.

4

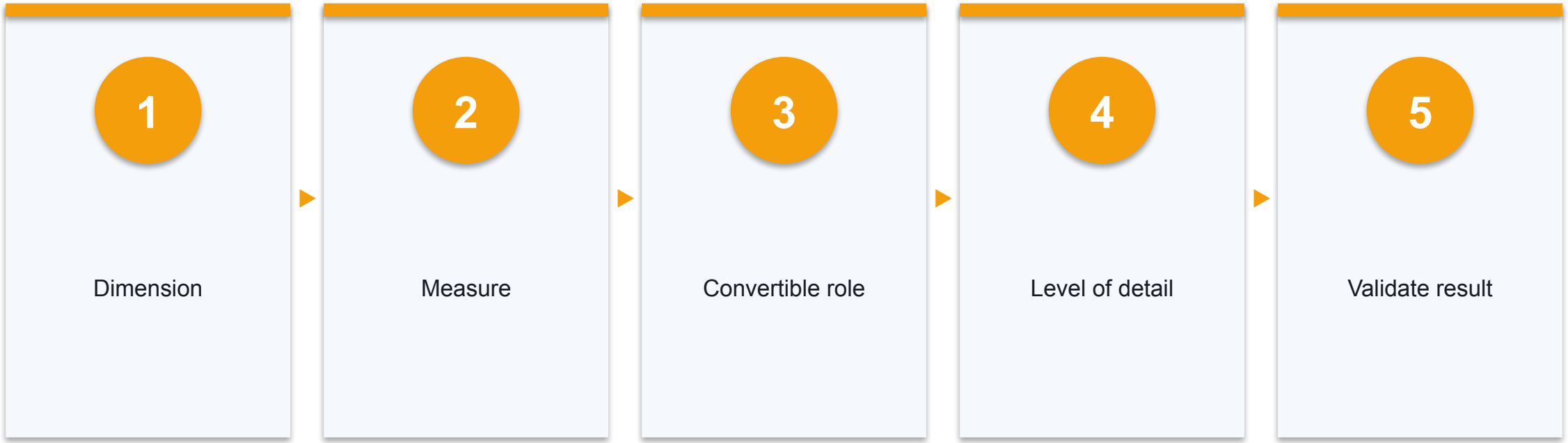
Level of detail

Every dimension changes the granularity of marks.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Understand dimensions and measures



This is a reasoning map, not a click-by-click lab procedure.

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4.1.1 · What dimensions contain

1

What it means

Dimensions usually contain names, categories, dates and locations.

2

Use it when

Use them to segment, label, filter and define detail.

3

Exam clue

They often create headers and partitions.

4

Common pitfall

A numeric code is not automatically a measure.

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4.1.2 · What measures contain

1

What it means

Measures usually contain quantitative values for aggregation.

2

Use it when

Use for totals, averages, counts, ratios and axes.

3

Exam clue

Green measures create axes; blue measures can create headers.

4

Common pitfall

Percentages and prices are not always additive.

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4.1.3 · Dimensions versus measures

1

What it means

Dimensions answer who/what/where/when; measures answer how much/how many.

2

Use it when

Use both to define grain and numerical result.

3

Exam clue

Converting role changes partitioning and aggregation.

4

Common pitfall

Do not confuse role with data type.

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Understand discrete and continuous fields

1

Discrete

Finite headers that partition the view; blue pills.

2

Continuous

Numeric or temporal range on an axis; green pills.

3

Independent of role

Dimensions and measures can each be discrete or continuous.

4

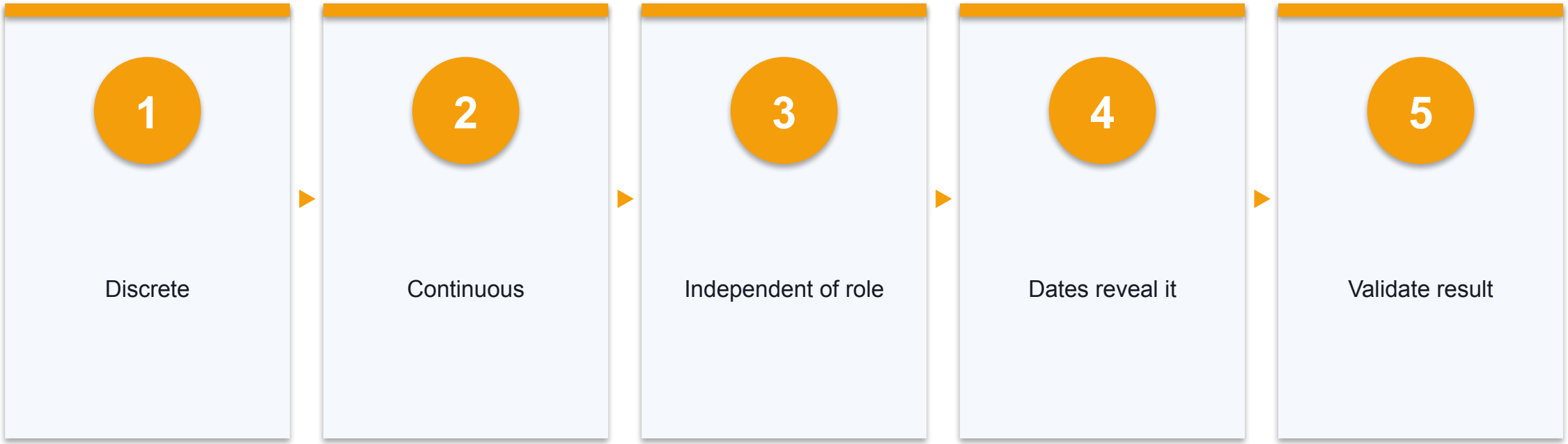
Dates reveal it

Date parts are commonly discrete; date values commonly continuous.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Understand discrete and continuous fields



This is a reasoning map, not a click-by-click lab procedure.

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4.2.1 · How discrete fields display

1

What it means

Discrete fields create headers and partitions.

2

Use it when

Use for named groups, panes and exact date parts.

3

Exam clue

Blue pills mean discrete behaviour.

4

Common pitfall

Discrete does not mean text.

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4.2.2 · How continuous fields display

1

What it means

Continuous fields create axes over an unbroken range.

2

Use it when

Use for quantitative scales and temporal trends.

3

Exam clue

Green pills mean continuous behaviour.

4

Common pitfall

Continuous does not mean measure.

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4.2.3 · Discrete date parts vs values

1

What it means

Date parts classify recurring members; date values preserve chronology.

2

Use it when

Use continuous dates for timelines and parts for seasonal grouping.

3

Exam clue

Blue MONTH shows Jan-Dec; green MONTH spans time.

4

Common pitfall

Month-only can aggregate multiple years together.

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Understand aggregations

1

Default behaviour

Numeric measures usually aggregate as SUM.

2

Grain changes result

Adding a dimension partitions rows into more marks.

3

Aggregation is contextual

The same measure differs at table, pane and cell levels.

4

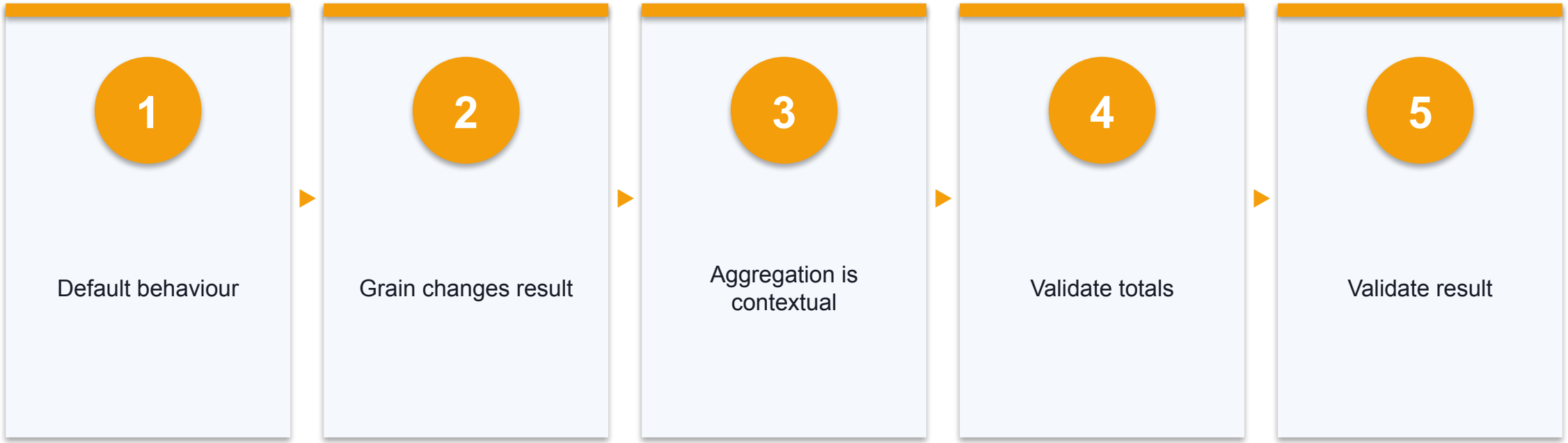
Validate totals

Ratios and distinct counts require suitable total logic.

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CONCEPTUAL DECISION FLOW

Reasoning Pattern — Understand aggregations



This is a reasoning map, not a click-by-click lab procedure.

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4.3.1 · Default aggregation

1

What it means

Measures aggregate to the current grain, commonly using SUM.

2

Use it when

Change to AVG, MIN, MAX, COUNT or another appropriate operation.

3

Exam clue

The pill shows the active aggregation, such as SUM(Sales).

4

Common pitfall

Number format and aggregation are separate.

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4.3.2 · How dimensions change aggregation

1

What it means

Each dimension partitions data into a finer detail.

2

Use it when

Use dimensions deliberately to answer the required breakdown.

3

Exam clue

SUM(Sales) by Region differs from Region and Customer.

4

Common pitfall

Unexpected totals may signal duplicated model rows or unintended dimensions.

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LAB 10 · TOPIC 04 · OVERVIEW ONLY

Capstone: Northstar Retail Executive Workbook

Northstar Retail's COO needs an executive workbook that reconciles sales, returns, inventory and regional targets, then communicates actions across desktop and mobile.

1. Model trusted data	2. Analyse questions	3. Build dashboard	4. Package insight	5. Review domains

OUTPUT A validated multi-table model, eight analytical worksheets, executive dashboard, phone layout, story, packaged workbook and exam-domain checklist

Detailed click-by-click procedure: Learner Guide and this lab folder · Tools: **Tableau Desktop · relationships · calculations · charts · dashboards · stories · TWBX**

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VERIFY THE OUTCOME

Lab 10 — Acceptance Test

✓ Evidence standard

Control totals reconcile, the model avoids duplicated measures, all eight worksheets and required analytics exist, dashboard/actions work on desktop and phone, exports are current, and every official objective has evidence.

Submit the Tableau workbook, requested screenshots/exports and completed acceptance-checklist.md from the lab folder.

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RECAP

Topic 04 — Exam Check

1

Explain

Define the key ideas in understanding tableau concepts and certification preparation.

2

Choose

Select the right feature from a short business scenario.

3

Predict

Reason about the result of shelf, model or formatting changes.

4

Prove

Use the lab workbook and acceptance evidence to validate the outcome.

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WRAP-UP

Certification Preparation & Review

Review the official blueprint, practise with the eleven workbooks, then demonstrate evidence

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COURSE SUMMARY

What You Achieved

1

Connect and prepare

Built trusted connections, extracts, models and metadata.

2

Explore and analyse

Created chart patterns, organisation structures and analytics.

3

Share insights

Built dashboards, actions, device layouts and sharing plans.

4

Understand concepts

Reasoned from field role, pill behaviour, grain and aggregation.

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THREE LAYERS

Certification Readiness

Know

- Explain every official objective
- Distinguish similar features
- Recognise common pitfalls

Do

- Complete all eleven workbooks
- Rebuild without instructions
- Validate totals and evidence

Perform

- Read the full scenario
- Eliminate answers by consequence
- Manage time across 40 items

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Exam-Day Reasoning

1. Identify the requested outcome
2. Name the Tableau layer
3. Predict the pill / grain / filter effect
4. Eliminate unsafe choices
5. Answer and move on

Official guide states 40 multiple-choice/multiple-select items and no outside materials.

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Official Preparation Resources

1. Read the complete Salesforce Certified Tableau Desktop Specialist exam guide and objective list.
2. Use Tableau's free Desktop training videos for connection, workspace, concepts, dashboards, mapping and calculations.
3. Repeat each lab with a clean workbook until the objective is automatic.
4. Use the Exam Domain Matrix in Lab 10 to locate weak areas.
5. Finish with the exact Tableau Desktop Specialist practice-exam link on the practice-exam slide.

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SALESFORCE CERTIFIED TABLEAU DESKTOP SPECIALIST

Official Certification Exam Details

40 + up to 5
MC / MS items

70 min
Time limit

48%
English pass score

USD 75
Registration fee



Connect & prepare



Explore & analyse



Share insights



Tableau concepts

No prerequisite · Product version 2022.3 · No hard-copy or online references · Retake fee: free

Always confirm registration details on Salesforce before booking; fees and policies can change.

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EXAMNOVA · ORIGINAL BLUEPRINT-ALIGNED PRACTICE

Tableau Desktop Specialist Practice Exam

SCAN TO START

3 × 65
questions

60 min
per set

70%
practice target

USD 20
practice tier

USD 100
voucher tier

Practice bundle domain mix

25% Connect & Prepare · 35% Explore & Analyse · 20% Share Insights · 20% Concepts

https://exams.tertiaryinfotech.com/practice-exams/tableau/tableau-desktop-specialist

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Course Feedback

1. Tell us what worked well and what could be clearer.
2. Use the feedback link or QR code shared by your trainer.
3. Rate the content, labs, pacing and trainer delivery.
4. Your comments help us improve future runs of this course.

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TABLEAU DESKTOP SPECIALIST

Thank you — build the evidence, not just the view.

A trusted Tableau insight connects the right data, uses the right grain, communicates clearly and survives validation.

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